



Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- (i) What is the role of knowledge in reasoning?
- (ii) What is a quantifier in predicate logic?
- (iii) Differentiate uniformed and informed search?
- (iv) What is the role of the environment in AI agent's decision-making?
- (v) What is the main objective of an Adversarial search?
- (vi) Explanation-based learning involves _____.
- (vii) What is the formal notation used to represent predicate logic statements?
- (viii) What is pragmatic processing?
- (ix) What is state space search in AI?
- (x) Which heuristic search strategy combines the advantages of both breadth-first search and depth-first search?
- (xi) What role does intuition play in reasoning?
- (xii) $(p \text{ OR } q) \text{ AND } \neg(p \text{ AND } q)$ is this a tautology?

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. What are logical connectives in predicate logic? [5]
3. What are some challenges in natural language processing? [5]
4. What are the advantages of the MYCIN? [5]
5. Convert the following into 1st-order predicate logic [5]
 - a. Someone walks and talks
 - b. Someone walks and someone talks
 - c. Everyone who walks is calm
6. Missionaries and Cannibals is a problem in which 3 missionaries and 3 cannibals want to cross from the left bank of a river to the right bank of the river. There is a boat on the left bank, but it only carries at most two people at a time (and can never cross with zero people). If cannibals ever outnumber missionaries on either bank, the cannibals will eat the missionaries. Find the solution to the problem. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. (a) What is Production System? [2]
 (b) What are the features of production system? [5]
 (c) What are the classes of production system? [5]
 (d) What are the advantages and disadvantages of production system? [3]
8. (a) Suppose you are the interrogator in Turing test. Compose five questions (giving logic behind them) that you would ask to determine which one is human and which is not. [5]
 (b) Define the following: Agent, agent function, agent program, rationality and reflex agent. [10]
9. (a) Explain Min-Max search procedure [6]
 (b) What is hill climbing algorithm. explain with an example. [6]
 (c) What are the limitations of hill climbing algorithm? [3]
10. (a) What are the quantifiers used in predicate logic? [5]
 (b) What is first-order predicate logic? Give an example. [5]
 (c) [5]

Express the following in FOL and construct a proof using resolution refutation:

1. Everyone has a parent
2. For any persons x, y, and z, if z is y's parent and y is x's parent, then z is the grandparent of x.
3. Therefore, everyone has a grandparent.

11. (a) What is Inheritable Knowledge? [5]
 (b) What is Inferential knowledge? [5]
 (c) Explain control knowledge and meta-knowledge. [5]